

Does AI really harm the environment as much as most people say it does?

Posted by u/thepooptrains • 0 points • 24 comments

I used to use AI obsessively in the past, however I mostly stopped after I found out that it destroys the environment. A lot of people say it harms it a lot, however, I've seen others say it's just over exaggerating so people hate on AI even more. Which claim is true?

Comments

u/Oreo97 • 1 points • 2025-06-17 10:16 UTC

It's not AI that's the issue it is power production which is primarily done with gas and coal power plants not clean energy sources like nuclear or wind (solar panels doesn't count because the panels are low efficiency, have short lifespans and can't be recycled).

u/rileystoneill • 1 points • 2025-06-17 23:13 UTC

Solar panels are cheap and last long enough. Nuclear power is very expensive and requires regular maintenance which is incredibly expensive.

u/Oreo97 • 1 points • 2025-06-18 03:23 UTC

Actually in the 60s it was proven nuclear power could be designed in a set and forget manner. Nuclear reactors have a design life of 50 years and a typical operational life of 60-75 years...

Wind turbines require constant maintenance, about a 3rd of your average nuclear plant can't be entered so maintenance is required to be minimal.

>Solar panels are cheap and last long enough.

Are they really? A lifespan of 15-20 year, a maximum efficiency of 25-30%, and they can't be recycled.

Lets put this in perspective by far the cleanest and safest forms of power production are wind, nuclear and solar which all kill less than 1 person per Terrawatt hour.

[our world in data](<https://ourworldindata.org/grapher/death-rates-from-energy-production-per-twh>)

So tell me what's cheaper? Permanent infrastructure that just needs to be maintained for generations to come or something that needs to be replaced every generation or

sooner.

Think about it this way, is it cheaper long-term for me to buy a Henson's razor or keep buying cartridge razors that barely survive one shave?

u/UnstableUnicorn666 • 1 points • 2025-06-17 09:47 UTC

There is no easy way to compare it. Just like saying that car would give this amount emissions, the other option is not to stay home, but compare it to train, or moving to city, or other options.

For example I just used AI to help me with API calls. Is it really more resource heavy to ask AI and get the answer and succeed at first try, than google it from multiple sources, try different calls that fail multiple times, until I find the way achieve my goal?

In this case I had my computer open 15 minutes and I visited 3 sites, one being the AI. Without AI, I would of google the API instructions, read those for 15minutes, googled some examples, tested, noted that way that I was thinking doing it does not work, google some more, test some more and maybe figure it out propably in few hours.

I don't say that AI does not do harm, and we should not look into it. But it's not as black and whote as some people make it out to be. We as humans should be more consious of the resources that we use and choose better options.

u/edgarallanbore • 2 points • 2025-06-17 17:49 UTC

Cutting hours of trial-and-error with one solid AI query saves both juice and sanity. I spin up Postman to test endpoints, let DreamFactoryAPI auto-generate the boilerplate, and lean on APIWrapper when I need bulk transformations; a single chat gets me the right curl sample, so my laptop isn't thrashing for three extra hours. You can push the savings further by batching prompts, caching docs offline, and killing tabs once you've got the answer. Sure, servers behind the model draw power, but so do all the dead-end searches we skip. Smarter use beats ditching AI altogether.

u/honeytwinklydream • 1 points • 2025-06-17 09:44 UTC

Yeah, it's definitely a mixed bag. AI does use a ton of energy, especially when training huge models, like the GPTs, image generators, etc. That part isn't really exaggerated. It takes massive data centers and computing power, and that means a big carbon footprint depending on how green the energy source is. So yeah, it's not "harmless" by any stretch. But I also think some people blow it out of proportion without comparing it to other tech or industries. Like, streaming video 24/7, gaming, crypto, even just driving around, they all burn energy too. It's more about how we use AI and whether companies are actually investing in sustainable solutions.

So IMO: yes, it has an impact, but it's not some world-ending monster unless we keep scaling it irresponsibly without pushing for clean energy.

u/tinny66666 • 1 points • 2025-06-17 09:17 UTC

It's true that AI uses a lot of power/water, but when people tell you this they mysteriously neglect to put that in context - this indicates they may have an agenda.

Data centers in general use a lot of power/water. Here's a comparison of water use for various online activities that does put it in context: [<https://x.com/SteveRattner/status/1884971579236401300>] (<https://x.com/SteveRattner/status/1884971579236401300>)

Water use is a proxy for electricity use since each watt of power used requires a proportional quantity of water to cool, so the graph of electricity use looks much the same, but with different units.

u/Geartheworld • 1 points • 2025-06-17 09:14 UTC

If you ask if it is bad for the environment, the answer is yes. But almost everything we are doing is bad for the environment. If the return is worth more than the environmental impact, I think we should still use it when necessary, but keep on finding a way to reduce the bad impact.

u/Suit_Difficult2525 • 1 points • 2025-06-17 09:02 UTC

AI's environmental impact isn't black & white. It's just another tool; the key is how we choose to use it. We need to balance innovation with stewardship.

u/ErusTenebre • 3 points • 2025-06-17 09:00 UTC

I think it's mostly focused on the concept of training models, not really each time you actually prompt a model.

Some models are so lightweight you can run them on a moderately strong PC.

You probably use more power/create more waste playing a multiplayer game like Fortnite.

u/gameryamen • 23 points • 2025-06-17 08:53 UTC

Like most cultural hot button issues, the truth is complicated. It's definitely true that training AI models is a costly process. It requires a lot of time for a large computing cluster to do, which means it takes a lot of power and requires a lot of cooling. That's why some companies (like Microsoft) are building new power plants, anticipating that this need for heavy computation training will continue over the next few decades.

Power isn't the only impact on the environment though. Servers have to be kept cool, they produce heat when they are working. This is typically done by systems that move water around to carry heat away from the servers. That heat has to go somewhere, and venting it back into the air (or dumping our warm water) has an environmental impact. Many modern data centers and super clusters are optimized to reduce this issue, by reusing water and dissipating heat in safer ways, but better isn't perfect.

On the other end, prompting an AI model and getting a response doesn't require much energy. You're not spinning up a whole supercluster when you send a prompt, just a single worker thread. I've seen estimates that a text prompt and response takes about the same energy as 5-8 Google searches.

If that prompt gets you a result you're satisfied with in one shot, it might be more efficient (energy wise) than doing a bunch of Google searches. But since LLMs hallucinate, we usually need to verify claims they make, so they actually create more Google searches.

u/Eastern-Zucchini6291 • 1 points • 2025-06-19 06:20 UTC

One thing I've seen is people lump in all data centers needs when talking about AI impact.

u/De_Mille • 6 points • 2025-06-17 09:14 UTC

The second problem is that google has implemented ai into its search engine. So if you dont want to use ai systems you need to install a different search engine. A google search now requires more energy as well.

u/chilling_guy • 2 points • 2025-06-17 10:06 UTC

Or search with "-ai" in your entry

u/BokChoyFantasy • -2 points • 2025-06-17 09:02 UTC

Thanks ChatGPT!

u/gameryamen • 1 points • 2025-06-17 16:33 UTC

That response wasn't written by any LLM.

u/Salt-Page1396 • -1 points • 2025-06-17 08:52 UTC

don't quote me on it but i heard somewhere that generating an ai image uses the equivalent power of charging a phone from 0-100%.

u/OfficialHashPanda • 1 points • 2025-06-17 10:26 UTC

It depends on the model, but that is indeed a pretty small amount of electricity.

u/EnderWigginson • -2 points • 2025-06-17 08:47 UTC

It consumes a lot of power, so in thst sense it's not good for the environment.

u/Inevitable-Regret411 • 2 points • 2025-06-17 08:45 UTC

There's plenty of credible news articles describing the environmental cost.

<https://www.theguardian.com/technology/article/2024/jul/02/google-ai-emissions>